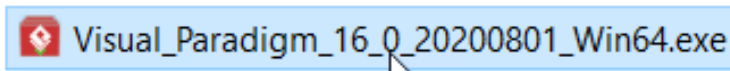


Lab 1: Getting Started with Visual Paradigm

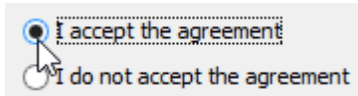
- Visual Paradigm is a powerful, and easy-to-use visual UML modeling, it provides software developers the innovative development platform to build quality applications faster and better.
- In this lab, we are going to use Visual Paradigm 16.0 to perform UML modeling. This software can be downloaded for from: <https://ap.visual-paradigm.com/qatar-university/license.jsp>

Installing Visual Paradigm

1. **Execute** the downloaded Visual Paradigm installer file. The setup wizard appears as below.



2. Click **Next** to proceed to the **License Agreement** page.
3. Read through the license agreement. If you fully understand and agree with them, choose **I accept the agreement** to accept the terms. Click **Next**.



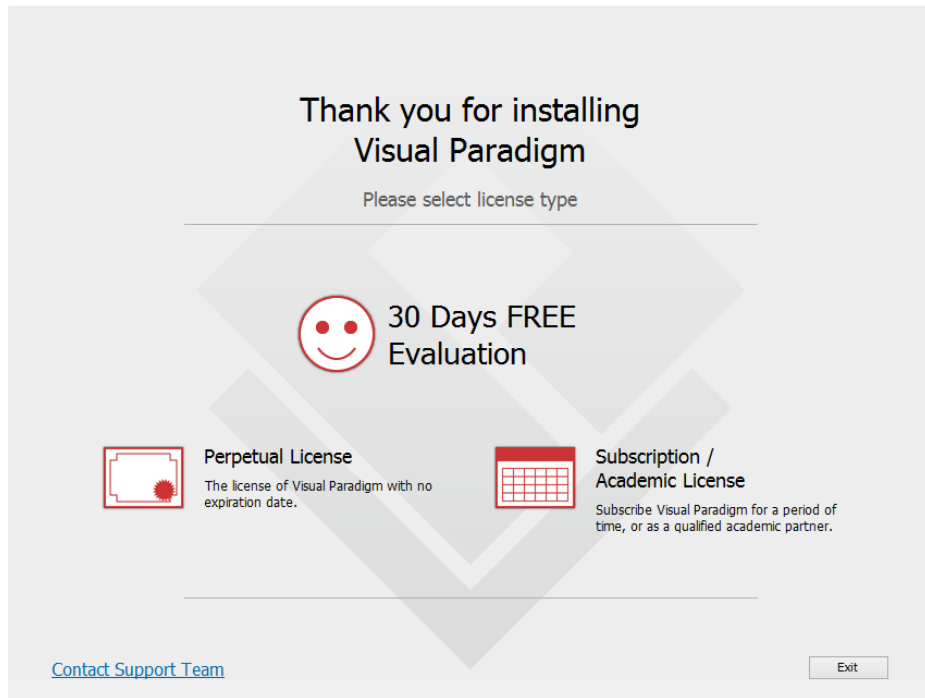
4. Specify the directory for installing Visual Paradigm. Click **Next** to continue.
5. Specify the name of the **Start Menu** folder that will be used to store the shortcuts. Keep **Create shortcuts for all users** checked if you want the shortcut(s) to be available in all the user accounts in the machine. Click **Next** to continue.
6. In the **Select File Associations** page, keep **Visual Paradigm Project (*.vpp)** checked if you want your system to be able to open the project file upon direct execution (i.e. double click). Click **Next** to start the file copying process.
7. Once the file copying is finished, you can choose to start Visual Paradigm immediately. On the other hand, you can finish the installation without starting Visual Paradigm. Choose the option **Don't Start** and click **Finish**. This ends the installation of Visual Paradigm.

Starting Visual Paradigm

Windows users can start Visual Paradigm via the Start or Start menu. If you selected not to create an entry in the Start menu (during the installation), you can look under the installation folder of Visual Paradigm (the same path specified in step 4 in the section above) and start Visual Paradigm by running **Visual Paradigm.exe** in the **bin** folder.

Selecting License Type

When you start Visual Paradigm the first time, you are asked to select a way to activate Visual Paradigm. The wizard appears as below.

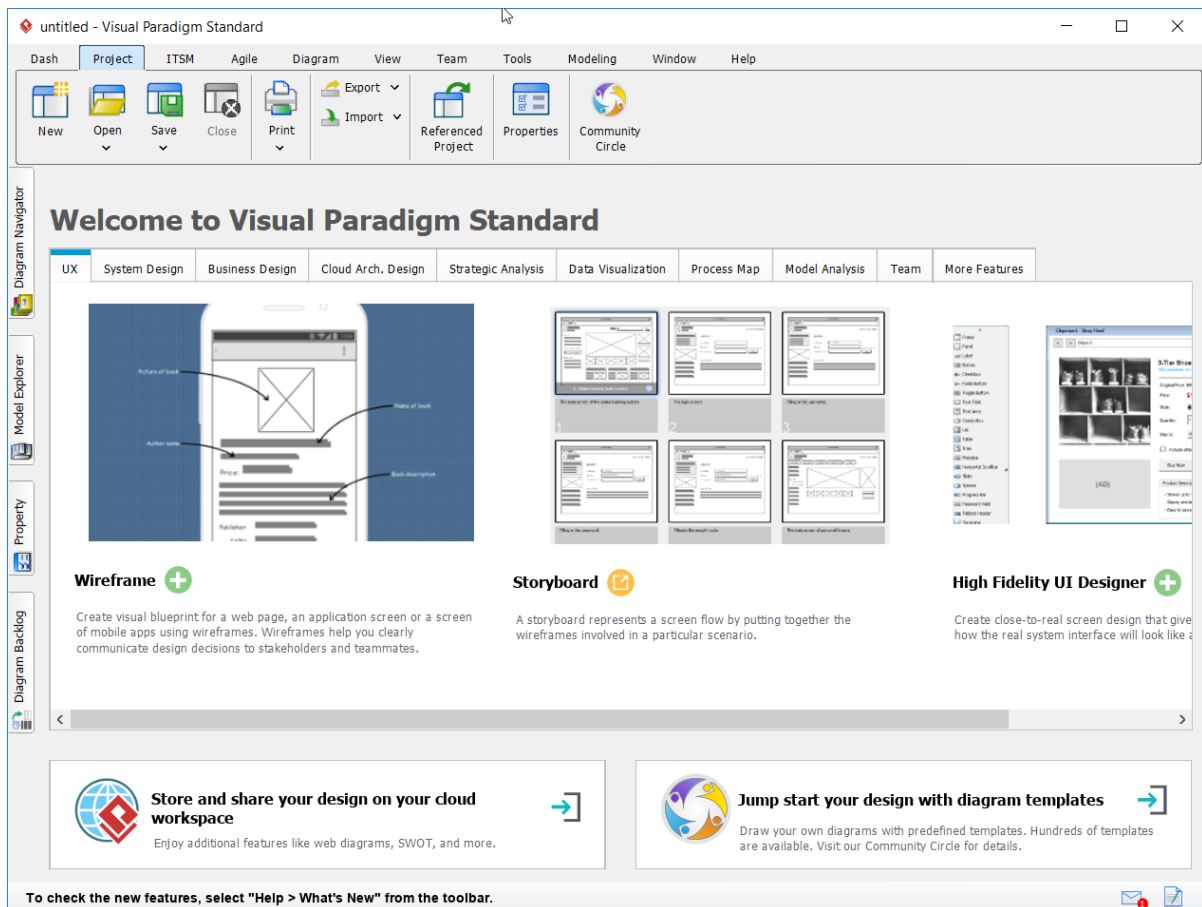


8. **Choose Subscription/Academic License**

9. Enter the activation code **XU2K6-T7SJ6-95J27-Y3ASL-YZ252** and click **Activate**. Note that Internet connection is required for verification.

10. You should receive a notification Email with an activation code. Copy the activation code first, and then click **OK** to continue.

11. The Visual Paradigm Environment appears as below.



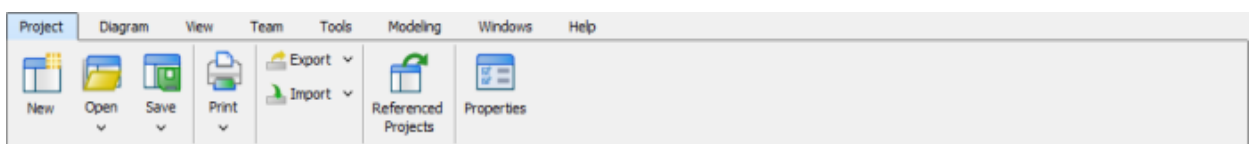
Visual Paradigm Interface

- Visual Paradigm 's user interface comprises a **toolbar**, a **diagram editor** and a **status bar**.

No.	Name	Description
1	Toolbar	A tabbed toolbar that allows you to perform various operations in Visual Paradigm.
2	Diagram editor	The diagram will be displayed in diagram editor.
3	Status bar	Notifications are shown here. You can also open the message pane and description pane from the bottom right of the status bar.

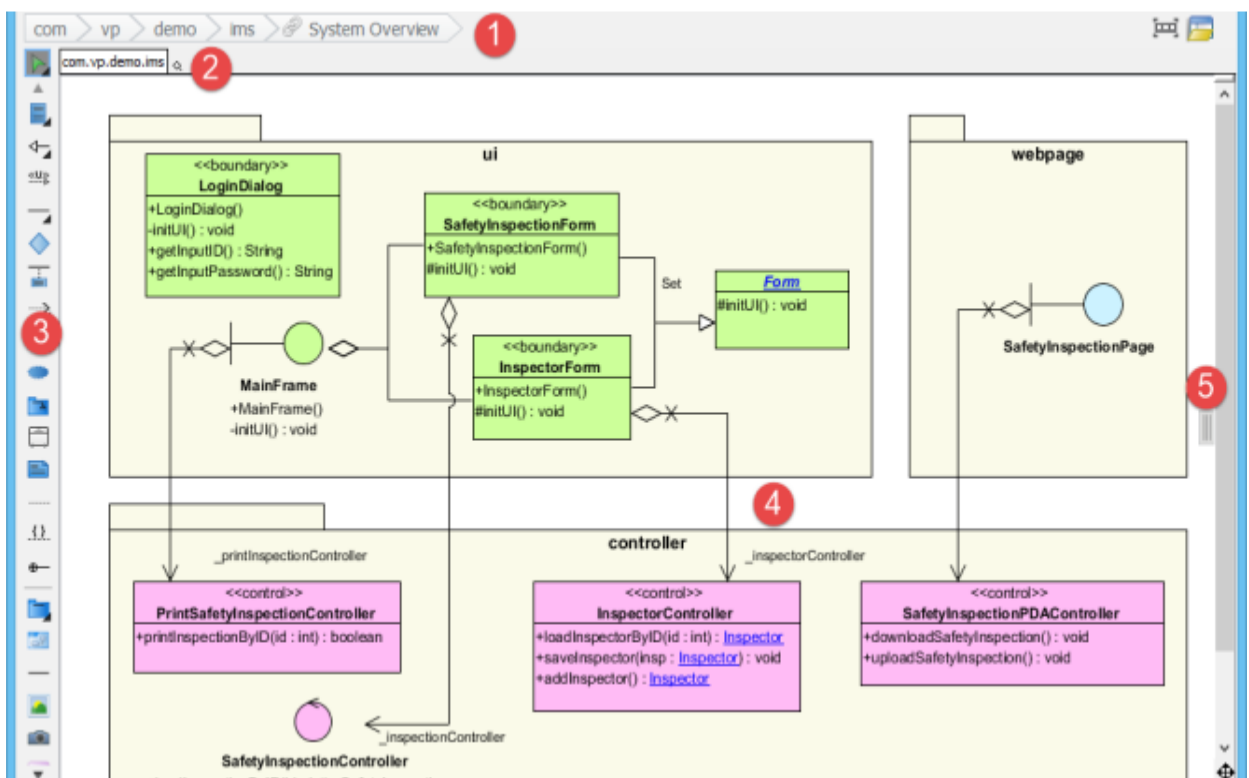
1. Toolbar

- The toolbar appears at the top of the Visual Paradigm application window. It is a collection of commonly used buttons and tools, categorized into several function tabs. The toolbar is shown by default, but you can collapse it by double clicking on any tab.



2. Diagram Editor

- Diagram editor is where you can view and edit your diagram.



Location	Name	Description
1	Navigation bar	Navigation bar consists of two parts. The left hand side contains a breadcrumb that shows the diagram name and the location of the diagram. The right hand side contains two buttons. One for switching to another diagram, another for opening the Project Browser.
2	Package header	The name of the package where the diagram resides. Package header is available only in some of the diagrams, but not all. You can double click on it to edit it. By entering the name of another package, the diagram will be moved to that package accordingly. You can click here to learn more about package header.
3	Diagram toolbar	The diagram toolbar provides you with the tools that you need to draw a diagram. Two kinds of tools can be found in the toolbar: - Diagramming utilities like the pan tool, sweeper and magnet tool. - Diagram -specific tools like actor in Use Case Diagram, BPMN pool in Business Process Diagram. For details about how to create a diagram with the toolbar, click here.
4	Diagram	The diagram to edit.
5	Action bar	Click to toggle the action bar that provides you with access to the tools based on your selection. For instance, if you have selected a Business Process Diagram, you can access tools like Working Procedure, Animation. If you have selected a use case, you can access tools like Use Case Details.

Description of diagram editor

3. Status Bar

- The status bar appears at the bottom of the application window. It reports events that occur when using Visual Paradigm.

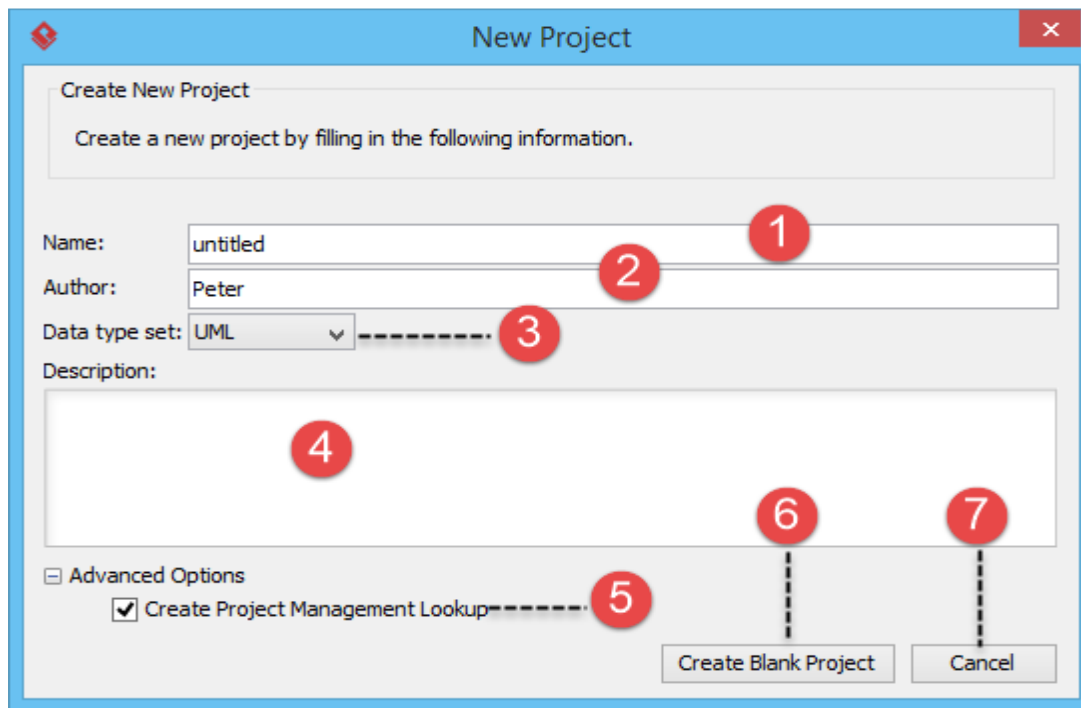


Creating Project

- Visual Paradigm stores information like model elements and diagrams in a project. Therefore, you need to create a project before performing modeling. To create a project, select **Project > New** from the toolbar. The New Project window appears.



- Visual Paradigm saves all project content to a single file, with file extension **.vpp** . To save your work, select either **Project > Save** or **Project > Save as**.



*An overview of New **Project** window*

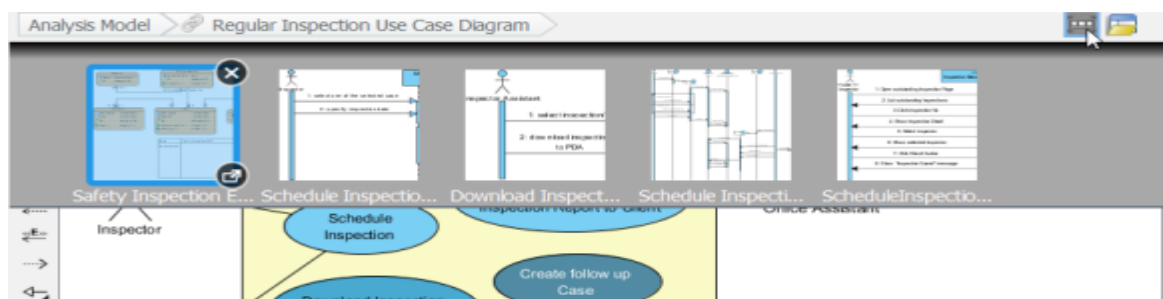
Project Browsing

- The Project Browser provides you with different perspectives in viewing your opening project. With Project Browser, you can view and open the diagrams in your project, browse model structure, check and open the recently modified diagram, etc.
- There are two ways you can take to open the Project Browser:
 - ✓ Select **View > Project Browser** from the toolbar
 - ✓ Click on **Open Project Browser on right hand side** of navigation bar of any diagram.



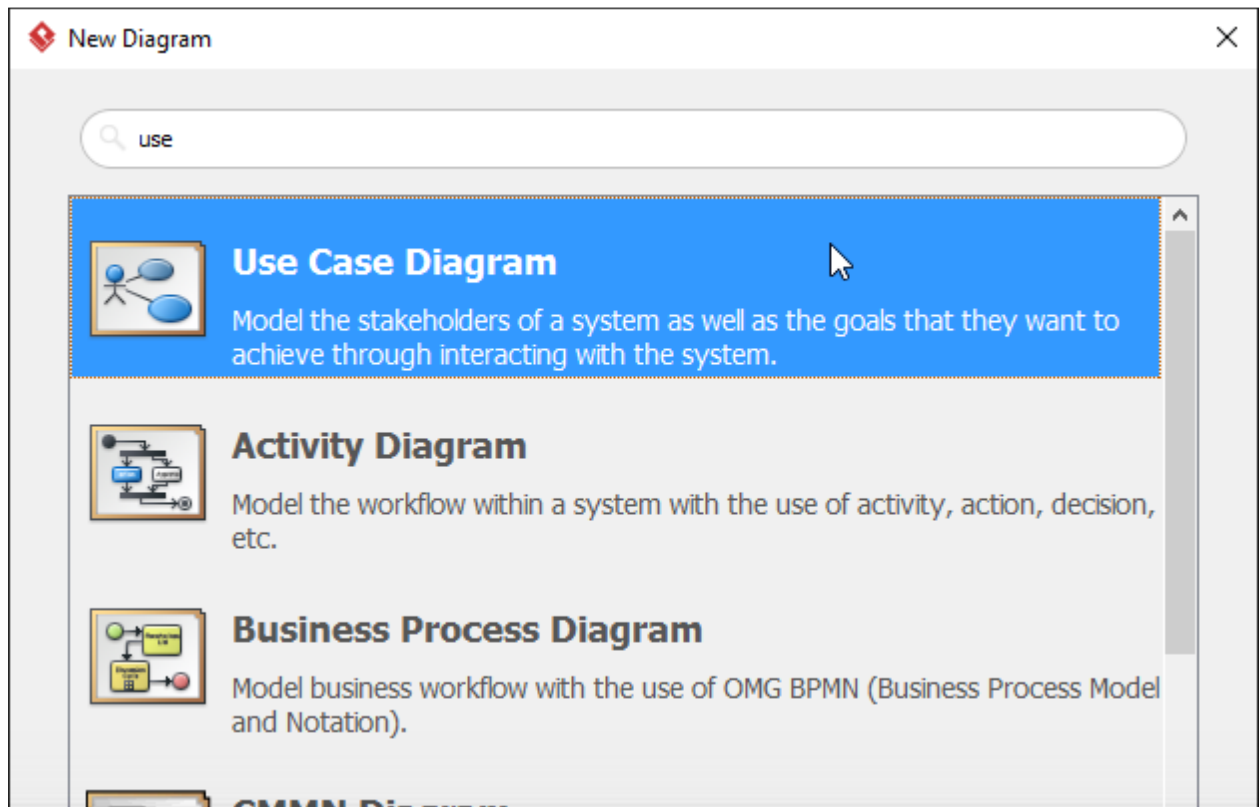
Switch Diagram

- After you've opened a few diagrams, you can switch to your target diagram easily with the feature of **Switch Diagram**. Moreover, you can close diagram(s) with this feature as well.
 - ✓ After you've opened a few diagrams, Click the **Switch Diagram button on the right** of the diagram pane.
 - ✓ Then, you can open a diagram by double **clicking on its thumbnail**.



Basic UML Diagramming

- This section will go through the steps of creating UML diagrams. To create a diagram:
 1. Select **Diagram** > **New** from the toolbar.
 2. In the New Diagram window, **select the type of diagram** to create.
 3. Click **Next**.
 4. Enter the **diagram name, its location and description**.
 5. Click **OK**.



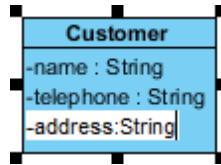
1. Class Diagram

- A class diagram models the blueprints of objects required by a system and the relationships between them.
- Let's make use of class diagram to model the domain classes of the **order processing system**.
- **Package Header:** When you create a class diagram, you are prompted to enter the package header (see the Figure below). Package header is a quick way to create new class diagram in a package. The diagram and the classes created in the diagram will all be grouped by the package specified. e.g. enter *mypackage* in package header.

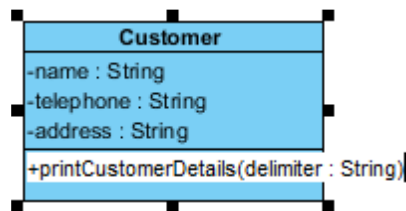


- **Attributes:** Properties of class are represented by attributes. To add an attribute:
 1. Right click on a class and select **Add** > **Attribute** from the popup menu.
 2. Type the name and type of attribute inline - for example, *name : String*.

3. Press **Enter** to confirm.
4. You can now create the next attribute by repeating step 2. To stop, press **Esc**.

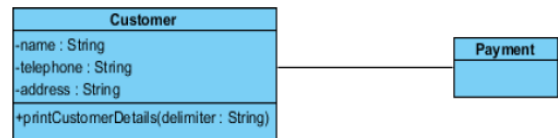


- **Operations:** Features of a class are represented by operations. To add an operation:
 1. Right click on a class and select **Add > Operation** from the popup menu.
 2. Type the name and optionally the method signature (i.e. parameters and return type) - for example, *printCustomerDetails(delimiter : String)*.
 3. Press **Enter** to confirm.
 4. You can now create the next operation by repeating step 2. To stop, press **Esc**.

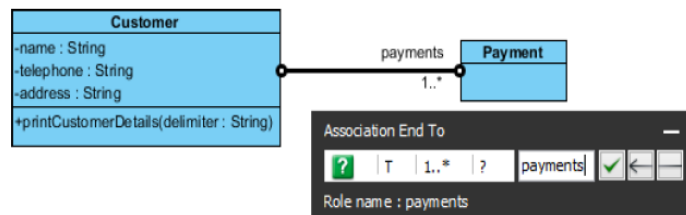


- **Classes with Association:** To associate two classes:

1. Move the mouse pointer over the source class.
2. Press on the **Association -> Class** resource icon and drag it out.
3. Release the mouse button to create a class with an association.
4. Name the class and press **Enter**.

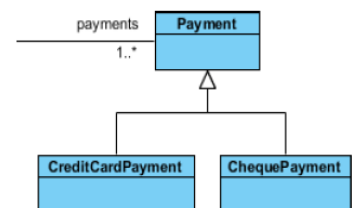


To edit an association end, double-click on the end to open the **Association Editor**. Enter a name for the role of the association in the middle text box and adjust properties like multiplicity and navigability as needed.



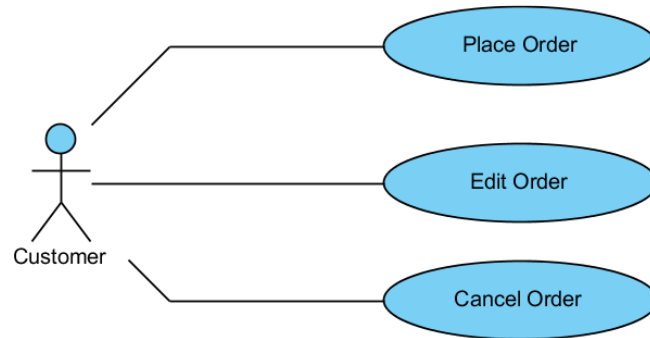
- **Classes with Generalization:** Generalization models “a-kind-of” relationship among classes. In Visual Paradigm, generalization can be created easily via the resource-centric interface, from super-class to sub-class.

1. Move the mouse pointer over the super class.
2. Press on the **Generalization -> Class** resource icon and drag it out.
3. Release the mouse button to create a class with a generalization.
4. Name the sub-class and press **Enter**.



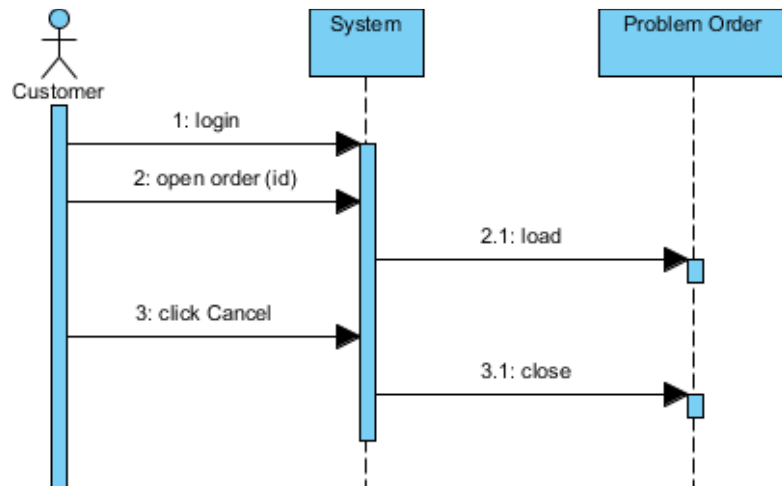
2. Use Case Diagram

- A use case diagram is used to model and identify the functional requirements of a software system. In a use case diagram, user and goals are represented by actor and use cases respectively.
- An actor is any person or external system that interacts with the system to achieve a user goal (i.e. use case).



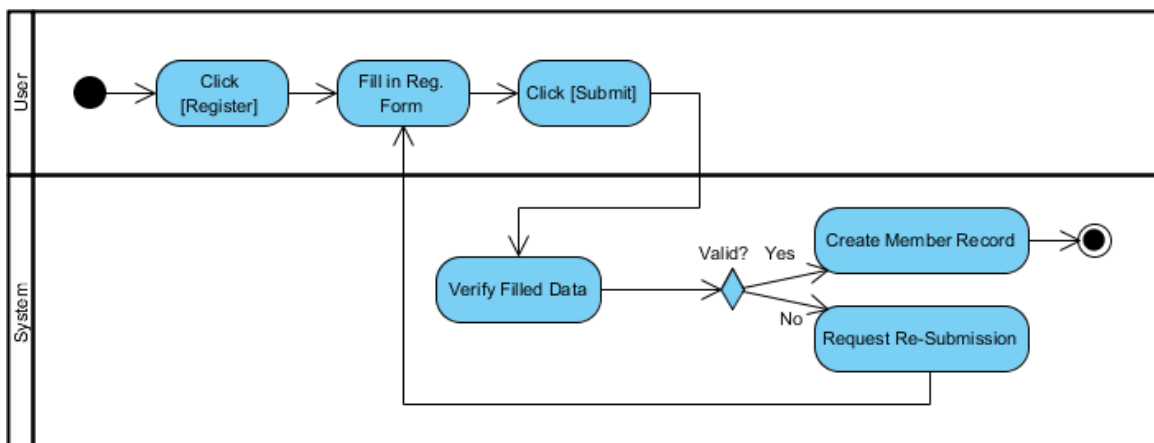
3. Sequence Diagram

- A sequence diagram is primarily used to show the interaction between objects that are represented as lifelines in a sequential order. More importantly, lifelines show all of their interaction points with other objects in events.



4. Activity Diagram

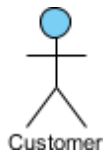
- An activity diagram is essentially a flowchart, showing the flow of control from one activity to another. i.e. it can model the dynamic aspects of a system.



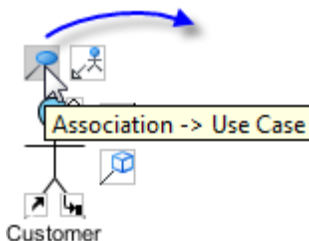
Exercise: Drawing Use Case Diagram in Visual Paradigm

In this exercise, we will make use of an **online hotel reservation** system as an example to demonstrate how to write effective use case with Visual Paradigm. Let's begin by drawing a use case diagram. We will carry on with writing effective use case with the resulting design.

1. Create a blank use case diagram by clicking UML in toolbar and selecting **Use Case Diagram** from the drop down menu.
2. Enter the name of diagram: **Hotel Reservation**.
3. Select **Actor** in the diagram toolbar. Click on the diagram to create an actor and name it **Customer**.



4. A customer can make a hotel reservation, which is a use case of the system. Let's create a use case from the Customer actor. Move the mouse pointer over the Customer actor. Press on the **Association -> Use Case** resource icon and drag it out.



5. Release the mouse button to create the use case. Name it **Make Reservation**. The association between actor and use case indicates that the actor will interact with the system to achieve the use case associated.
6. Complete the design to make it look like this:

